

OQI HACKATHON 2026 · BUAP · PUEBLA · EQUIPO 04

QFalcon — Resilient Release Scheduling for the International Falcon Reservoir

Classical & quantum optimization for transboundary water management · Río Grande / Río Bravo Basin

SDG 6.4 — Water-use efficiency

SDG 6.5 — Transboundary cooperation

SDG 13.1 — Climate resilience

26

weeks optimized (official)

+0.021

Δ SRS vs. historical (feasible)

0.02 s

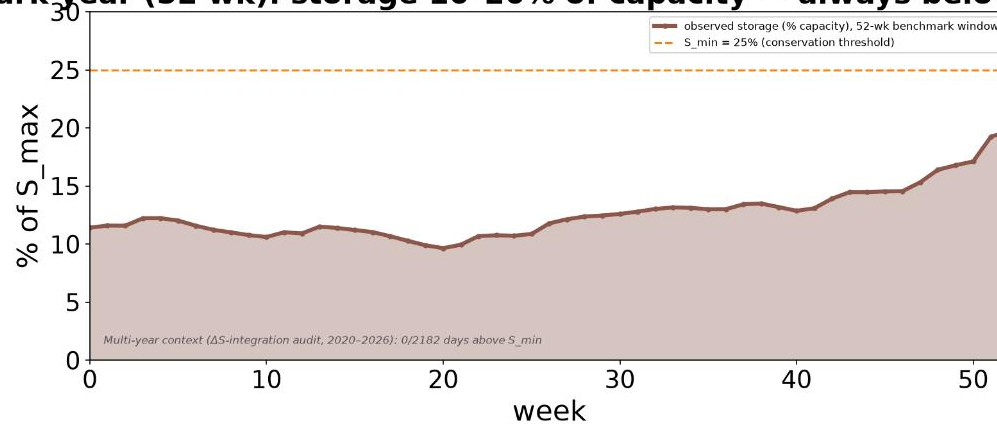
exact DP runtime

WHY IT MATTERS

A shared river under multi-year drought

The Falcon Reservoir sat below its 25% conservation threshold **every day of a 6-year record** — and its release timing affects communities on **both sides of the border**.

Benchmark year (52 wk): storage 10-20% of capacity — always below $S_{\min}$ (25%)



Scarcity. Storage ~10–20% of capacity; 0 / 2182 days above $S_{\min}$.

Cooperation. A US–Mexico treaty governs releases — SDG 6.5.

Resilience. Better timing buffers drought & climate stress — SDG 6.4 · 13.1.

PROBLEM FORMULATION

Optimize the weekly release adjustment $u(t)$

$$R(t) = R_{\text{obs}}(t) + u(t) \quad (u=0 = \text{historical}) \cdot$$

$$\text{SRS} = - (w_1 \cdot C_{\text{crit}} + w_2 \cdot C_{\text{dev}} + w_3 \cdot C_{\text{smooth}})$$

C_{crit}

penalizes weeks with $S(t) < S_{\text{min}}$ (25% of S_{max}) → prevents depletion

C_{dev}

penalizes deviation from historical releases → operational realism

C_{smooth}

penalizes abrupt week-to-week change → deliverable policy

Constraints: $R(t) \geq 0 \cdot |u(t)| \leq u_{\text{max}} \cdot 0 \leq S(t) \leq S_{\text{max}} \cdot \text{cumulative balance } |\Sigma u| \leq \eta \cdot \Sigma R_{\text{obs}}$.

DATA & OFFICIAL PARAMETERS

Official IBWC data, independently validated

$S_{\max}$ (official IBWC) $3.289 \times 10^9 \text{ m}^3$

$S_{\min} = 0.25 \cdot S_{\max}$ $822 \times 10^6 \text{ m}^3$

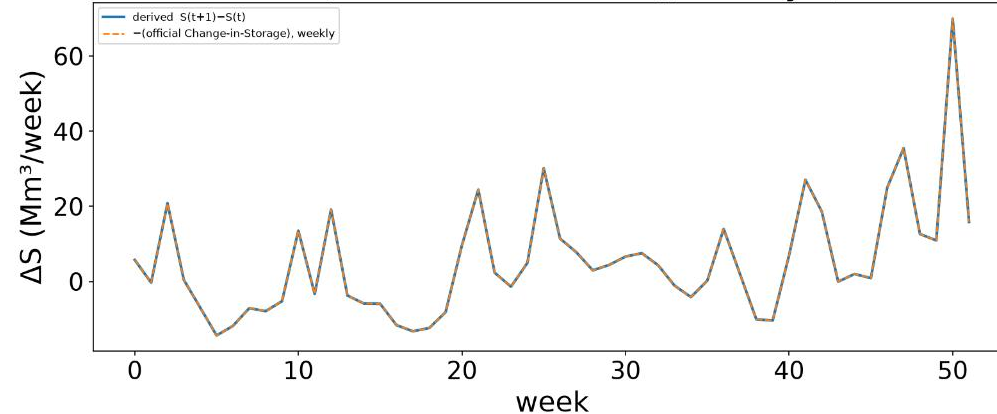
S_0 initial storage **$375.6 \text{ Mm}^3 \cdot 11.4\%$**

$\Delta u = 0.25 \cdot \text{median}(R)$ $2.80 \text{ Mm}^3/\text{wk}$

η balance budget 0.10

$S_0 = 11.4\% < 25\% \rightarrow$ the reservoir **starts in the critical zone.**

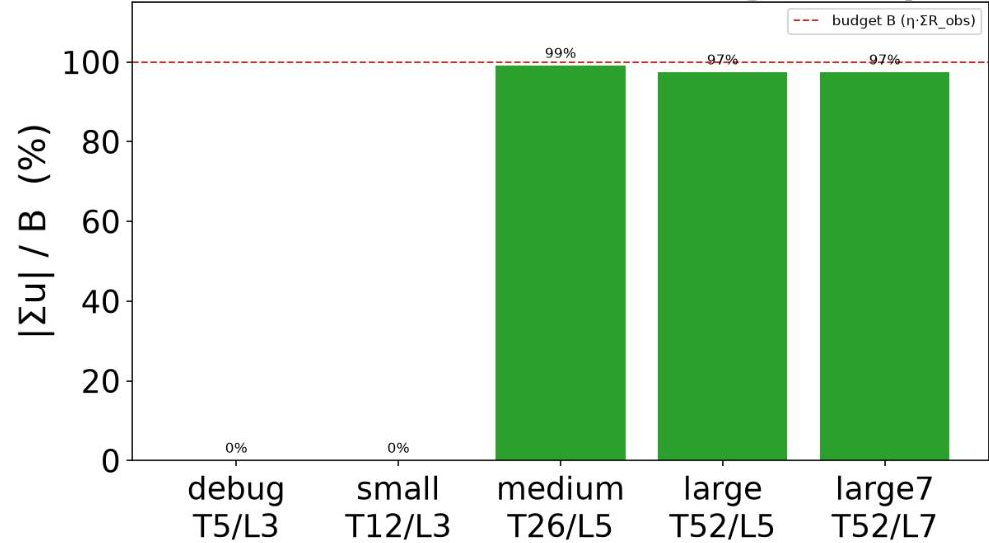
Derived $\Delta S == -(\text{official IBWC series}), \text{ weekly (corr}=+1.000)$



Every relaxation is measured, not assumed

We keep only the constraints that **actually bind** in the data — so the QUBO is small and **provably lossless**.

Balance constraint is (near-)binding → keep it EXACT



Per-constraint relaxation decision

constraint	binds?	relaxation	qubits
$0 \leq S \leq S_{\max}$	never	drop	0
$ \Sigma u \leq B$	~100%*	exact 1-slack	$\sim \log_2(0.8T)$
$R(t) \geq 0$	sparse	prohibit $ v $	0
one-hot $\Sigma=1$	always	penalty / XY	0

Storage bounds: never bind → dropped (0 qubits).

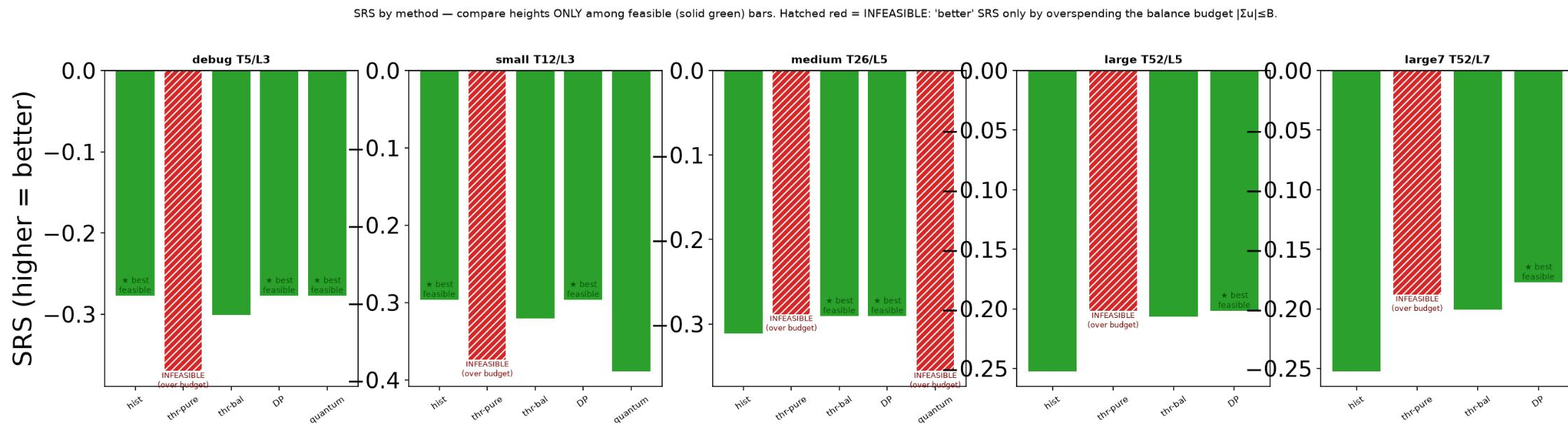
Balance $|\Sigma u| \leq B$: binds ~97–99% → exact 1-slack.

$R \geq 0$ / one-hot: forbidden-level penalty / XY-mixer.

BASELINE

Compare only among **feasible** methods

A taller bar can be **infeasible** — "better" SRS only by overspending the balance budget. **Read the green bars**; exact **DP** is the honest strong baseline.

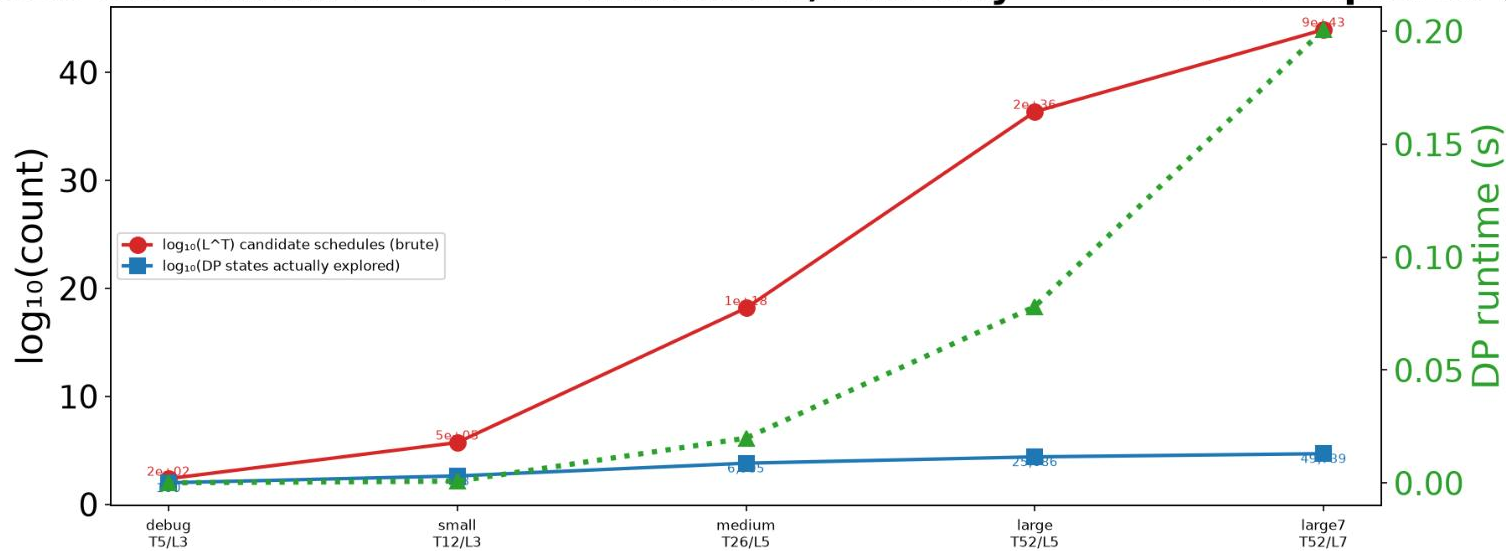


CLASSICAL BASELINE — EXACT DYNAMIC PROGRAMMING

10^{43} schedules $\rightarrow \sim 10^4$ states, solved exactly

Storage depends only on the integer cumulative sum $C_t = \sum k_j$, so exponentially many schedules **fuse** into an $O(T^2 \cdot L^2)$ state lattice — exact and lossless.

DP collapses the search: $L^T \rightarrow 10^{43}$ candidates, but only $\sim 10^4$ states explored ($O(T^2 L^2)$)



Medium: 1.49×10^{18} candidates
 $\rightarrow 6,505$ states, **0.02 s.**

Large L7: $8.8 \times 10^{43} \rightarrow 49,739$
states, 0.20 s.

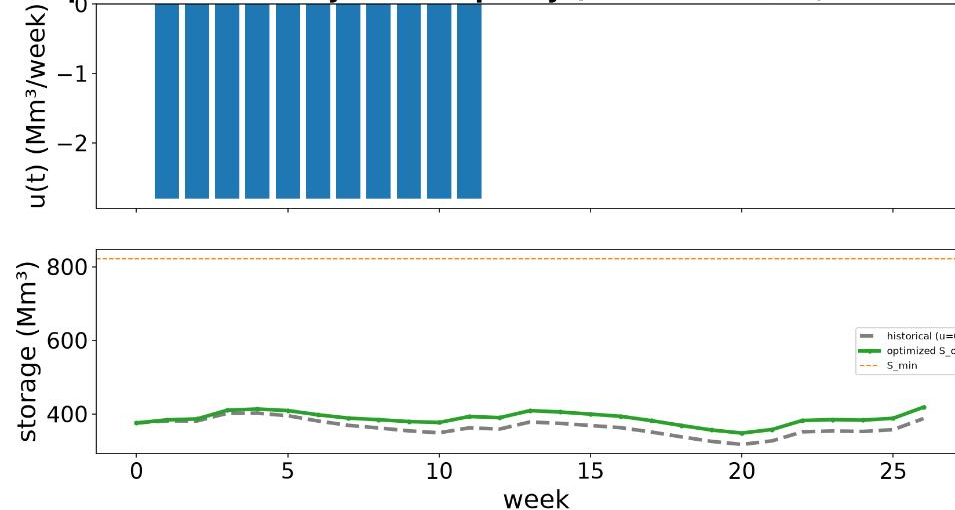
Verified: brute-force == DP on
small (531,441).

RESULT

What the optimizer **actually** does

Conserve ($u = -2\Delta u$) in **~11 of 26 weeks**, else hold historical — **redistributing releases in time**, not just withholding water.

DP optimal release-adjustment policy (medium T26/L5, $\Delta\text{SRS} = +0.0211$)



Lifts storage while spending 99% of the $\pm 31 \text{ Mm}^3$ balance budget.

Min storage 9.65% $\rightarrow$ 10.59% of capacity.

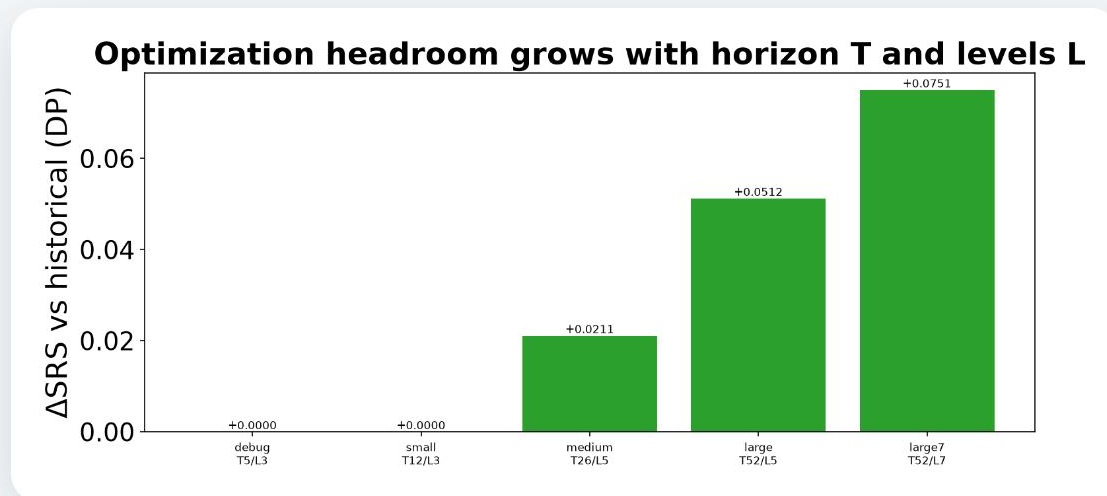
Feasible, $\Delta\text{SRS} = +0.021$ vs. historical.

BENCHMARKING

Official comparison & headroom

Method · T26/L5	SRS	Δ SRS	feas.
Historical (u=0)	-0.31153	+0.0000	✓
Threshold (literal)	-0.28893	+0.0226	✗
DP (exact)	-0.29042	+0.0211	✓
Sim. annealing	-0.29119	+0.0203	✓

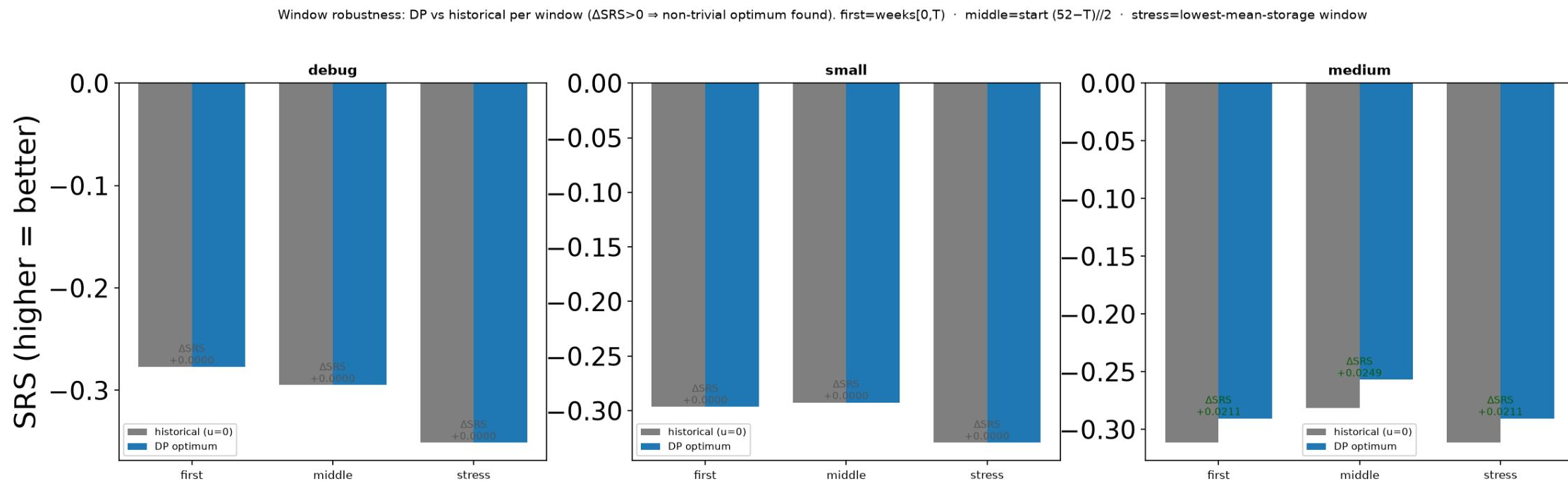
DP = balanced threshold exactly. SA needs ~330× the runtime and is slightly worse.



BENCHMARKING — ROBUSTNESS

Not an artifact of one time window

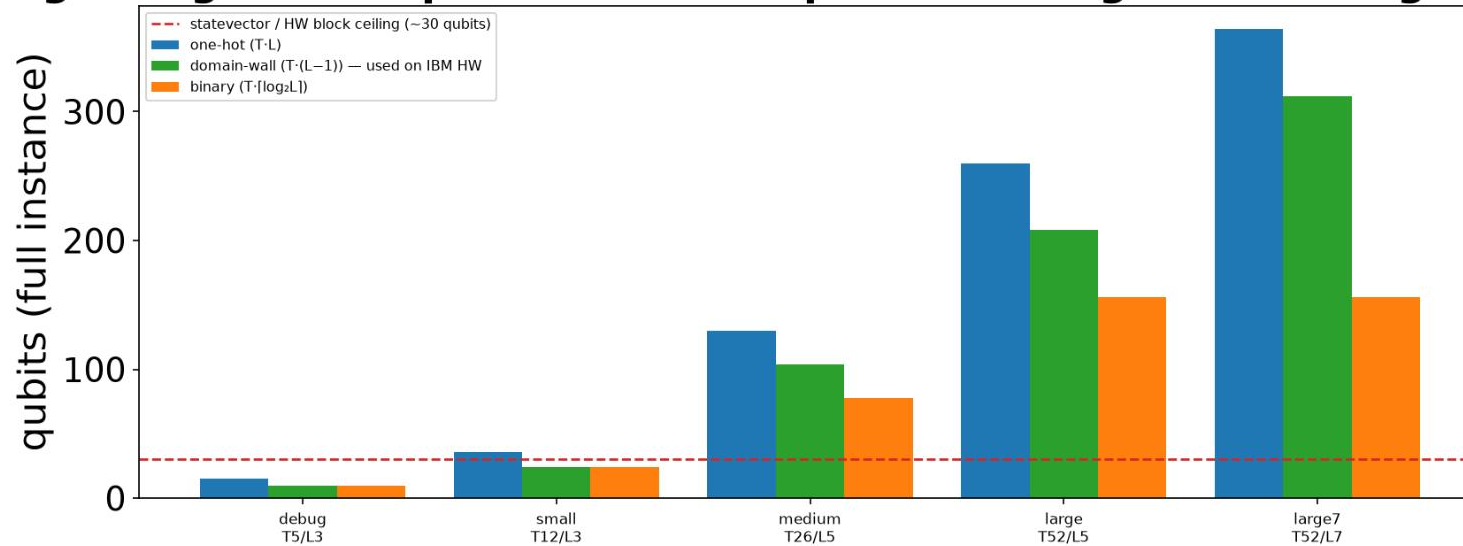
Re-run over 3 windows (first · middle · lowest-storage stress): medium beats historical in **all three**; debug/small stay $u=0$ — the result is **consistent**, not a fluke of week 0.



QUBO / QAOA — encoding the schedule

Every full instance exceeds ~ 30 qubits, so **compact encodings + temporal chunking** are necessary. Domain-wall is the encoding we ran on IBM hardware.

Encoding sizing vs ~ 30 -qubit limit $\rightarrow$ compact encodings + chunking are necessary



One-hot T·L; C_{dev}/C_{smooth} exact, balance 1-slack.

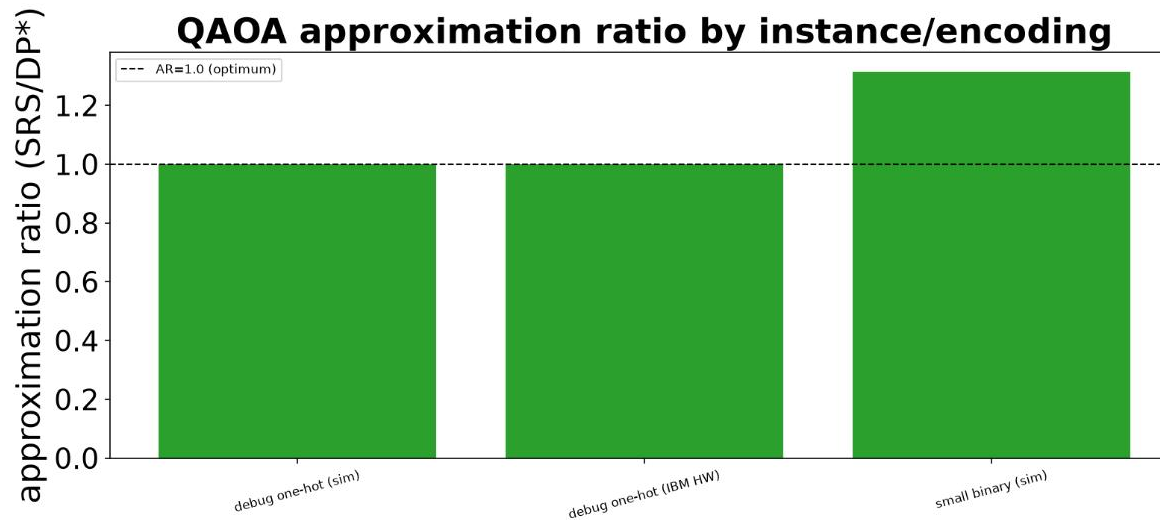
Binary halves qubits (T12/L3: 36 $\rightarrow$ 24, fits statevector).

Domain-wall 4 bits/week $\rightarrow$ IBM Quantum, 7+7+7+5 blocks.

QUANTUM TRACK — RESULTS

Exact on **real hardware** at small scale, honest at scale

Debug T5/L3 reaches the exact optimum (**AR = 1.000**) on the simulator **and real IBM hardware**. The **XY-mixer** keeps 100% of probability one-hot-valid, making chunked blocks feasible.



Medium T26/L5: solved by chunking, 25 qubits/block.

Gap to full-DP = chunking + QAOA depth (separated).

Honest: no quantum advantage yet — expected (spec §7).

CONCLUSIONS

Best feasible $\Delta\text{SRS} = +0.021$ — a 6.8% cut in penalized cost

Formulation

26 weeks $\times$ 5 levels; SRS with 3 penalties and 4 constraints — every relaxation measured, not assumed.

Classical optimum

Exact DP, verified by brute force; conserve 11 weeks then hold. Runtime 0.02 s.

Scalability

$10^{18} \rightarrow 10^{43}$ candidate schedules solved in < 1.1 s via cumulative-state reduction.

Quantum & limits

QAOA exact on hardware at debug; chunked + XY-mixer at medium. No advantage yet; simplified model (no treaty/legal terms).

IMPACT & ACKNOWLEDGEMENTS

Timing water better, under scarcity, together.

A reproducible benchmark for resilient release scheduling on a shared, drought-stressed reservoir — bridging classical exactness and quantum encodings toward SDG 6.4, 6.5, 13.1.

Use of AI

Claude (Anthropic) assisted with QUBO derivation, code scaffolding and documentation. All algorithms, benchmark decisions and results were designed and verified by the team.